Can LLMs Correct Historical OCR Without Rewriting History?

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A look at our survey, ***OCR Post-Correction in the Age of LLMs: A Survey of Historical Document Processing****.*

Authors: Emanuela Boros, Ahmed Hamdi, Wenjun Sun, Tien-Nam Nguyen, Mickaël Coustaty, and Antoine Doucet.

Read the full paper on HAL

Why historical OCR still needs correction

Digitised newspapers, books, and archival records make historical collections searchable. But their usefulness depends on the quality of the text extracted through optical character recognition (OCR).

Historical documents are particularly difficult: damaged pages, unusual typefaces, complex layouts, historical spelling, and rare vocabulary can all produce recognition errors. When those errors affect names, places, or dates, they can also disrupt search and information extraction.

Large language models (LLMs) and vision-language models (VLMs) offer new ways to recover this text. Yet a correction that reads smoothly may still change what the source actually says.

Our survey examines generative approaches published between 2022 and 2026, bringing together research on **methods, data resources, evaluation, and reliability**.

Four families of correction approaches

We organise the literature around four main approaches. These can overlap within a single pipeline.

Approach	How it works	A key consideration
Prompted text-only correction	An LLM receives noisy OCR and instructions, sometimes with examples.	Results depend on prompts, context, and the document collection.
Instruction tuning and fine-tuning	A model is adapted to correction tasks using training examples.	Training data must represent the errors and documents encountered in practice.
Controlled and reference-guided correction	Additional evidence or decoding constraints guide the correction.	Reliable references and suitable constraints are needed to control edits.
Multimodal correction	A model receives both the OCR text and the document image.	Access to an image does not automatically guarantee visually grounded output.

Direct image-to-text transcription is a related alternative, but it is a recognition task rather than post-correction: it generates a transcription without revising existing OCR.

No single approach consistently wins

Across the reviewed literature, generative methods can improve noisy OCR, but there is no universally dominant strategy.

Performance varies with language, document type, initial OCR quality, training data, model configuration, and available textual or visual context. Strong results on one collection do not necessarily transfer to another.

For practitioners, this makes evaluation on representative documents essential. Model size or general language ability alone does not establish suitability for historical transcription.

Better readability is not enough

Historical OCR correction aims to recover the source, including its unfamiliar or unusual forms.

Generative models may modernise historical spelling, change rare names, remove uncertain passages, or introduce plausible text unsupported by the document. Such outputs can be difficult to detect precisely because they are fluent.

Source fidelity must therefore be an explicit evaluation target.

What should we measure?

Character error rate (CER) and word error rate (WER) remain important measures of transcription accuracy. But they capture only part of the behaviour of generative correction systems.

Our survey highlights the need to assess several complementary dimensions:

- **Harmful edits:** does the model change text that was already correct?
- **Historical preservation:** does it retain valid historical spelling and rare forms?
- **Unsupported generation:** does it add content absent from the source?

- **Visual grounding:** where images are available, is the output supported by the page?
- **Structural fidelity:** are layout, reading order, and alignment preserved?
- **Downstream usefulness:** do corrections improve retrieval or information extraction?

Reproducibility also requires reporting prompts, model versions, decoding settings, preprocessing, and normalisation policies.

Data remains a bottleneck

Available resources differ in language, historical period, document genre, granularity, and supervision. The evidence remains concentrated on European historical newspapers and comparatively well-resourced languages.

Real OCR–ground-truth pairs provide realistic supervision but are expensive to construct. Synthetic errors and model-generated targets can increase scale, while introducing mismatches or biases that need to be evaluated.

Broader multilingual resources and more diverse archival document types are priorities for the field.

Correct existing OCR, or recognise the page again?

Modern VLMs make it increasingly relevant to compare three options:

1. Correct legacy OCR using text alone.
2. Correct it using both the OCR and the page image.
3. Produce a new transcription directly from the image.

The choice depends on image availability, legacy OCR quality, language, layout, computational cost, and how the resulting text will be integrated into the collection.

Our survey calls for systematic comparisons that examine accuracy alongside source fidelity, alignment, and operational effort.

Join the discussion

Reliable OCR correction can make historical material more accessible to search and analysis. The challenge is to recover that information while preserving the documentary evidence.

How do you check that an OCR correction improves your collection without changing the source? We would be interested to hear about evaluation practices, datasets, and failure cases from the community.

📄 Read the survey

Citation

Emanuela Boros, Ahmed Hamdi, Wenjun Sun, Tien-Nam Nguyen, Mickaël Coustaty, and Antoine Doucet. *OCR Post-Correction in the Age of LLMs: A Survey of Historical Document Processing*. 2026. HAL: hal-05756997.

👏 Community

stefan-it 17 days ago • edited 17 days ago

Hey @emanuelaboros ,

very interesting topic and super interesting research question. I thought a bit about it, here's some suggestion.

Main idea is to build first a dataset that consists of sentence variants, labels and an impact description:

Label	OCR	GT (if available)	Corrected	Impact
fixed	Laud	Land	Land	improved
missed	Kricg	Krieg	Kricg	neutral
modernized	Thür, ſeyn	Thür, ſeyn	Tür, sein	unfaithful
archaized	Haus	Haus	Hauß	unfaithful
wrong fix (real word)	Wcin	Wein	Bein	harmful
number changed	1o00	1000	100	harmful
entity damaged	Schmidt	Schmidt	Schmitt	harmful
hallucinated insertion	…der König von	…der König von	…der König von Preußen	harmful
omission	ſagte er, u. ſ. w.	ſagte er, u. ſ. w.	sagte er.	harmful
paraphrase	allwo er ſtarb	allwo er ſtarb	wo er starb	unfaithful
structural	Regie- / rung	Regie- / rung	Regierung	policy-dependent

As it can be seen a potential LLM correction from "1000" to "100" is super bad.

Impact labels are:

Impact	Meaning
improved	The correction moves the text closer to the source.
neutral	Nothing changed, or the change makes no difference.
unfaithful	The meaning is kept, but the historical form of the source is changed.
harmful	The content changes: different words, numbers or names, or text added or lost.

Impact	Meaning
policy-dependent	Whether the change is acceptable depends on the transcription guideline (diplomatic vs. normalized).

When an example contains several edits, its overall impact is the worst one, in this order: harmful > unfaithful > improved > neutral.

It is just my initial draft and ideas about potential labels.

I think one could construct a dataset from various input sources and train very good classifiers for it (different architectures can be possible: encoder, encoder-decodery, decoder-only and they could be a good step towards the intial "How do you check that an OCR correction improves your collection without changing the source?" question.

Reply

emanuelaboros Article author 17 days ago

Hello @stefan-it, glad to see your comment. I'm curious, the impact levels, are relative to what/whom? How could we/one automate such labelling?

Reply

stefan-it 17 days ago • edited 17 days ago

So the "impact levels" can be seen as some kind of coarse vs. fine-grained labels:

```
improved: fixed
neutral: missed
unfaithful: modernized, archaized, paraphrase
harmful: wrong fix (real word), number changed, entity damaged, insert
```

They can be assigned after fine-grained labelling, depending on the guidlines.

If you have OCR, ground-truth and the (LLM-)corrected sentences, then all three could be aligned at character-level using Levensthein alignment and differences can be grouped into edits. Each edit group can then be labelled (automatically), e.g. when the ground-truth edit group contains digits, then a number was changed or when text in extracted has no counterpart in OCR/ground-truth then it is an insertion, when text is missing from

corrected -> omission. This can be done automatically and would not require an LLM for labelling.

And later - instead of classifying a whole sentence, it is also possible to perform some span-labelling approach:

```
OCR:   d a s ␣ L a u d ␣ v e r w ü ſ t e t ␣ v o n ∅ ∅ ∅ ∅ ∅ ∅ ∅ ∅
Corr:  d a s ␣ L a n d ␣ v e r w ü s t e t ␣ v o n ␣ P r e u ß e n
Tag:   O O O O O O B-rep O O O O O O O B-mod O O O O O O B-ins I-ins I
```

-> It is possible to visualize which parts of a corrected sentence includes possible insertions (in this case it is a potential hallucination).

Reply

emanuelaboros Article author 12 days ago

Hello @stefan-it, I'll do some small experiments to see how it works out and come back to you!

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